

SATURATING THE COMPRESSION DEFECT: A TORSION-FREE FINITELY PRESENTED GROUP WITH NO SOFIC AND NO MF QUOTIENTS

DRAFT NOTE FOR EXPERT REVIEW — NOT FOR CITATION

ABSTRACT. We combine the construction of Fournier-Facio’s torsion-free non-sofic group with one further application of Hull’s small-cancellation theorem for acylindrically hyperbolic groups, routed so that the embedded simple factor normally generates the resulting quotient. The output is a 2-generated, finitely presented, torsion-free group Q with property (T), acylindrically hyperbolic, such that *every* nontrivial quotient of Q — including Q itself — is non-sofic and has full MF radical: every homomorphism from a nontrivial quotient of Q to the unitary group of a norm matrix corona is trivial. The two obstruction halves (the norm-corona radical theorem and the nonsocicity criterion) and all algebraic packaging are machine-checked in Lean; the geometric small-cancellation existence input is cited from Hull, Osin, and Fournier-Facio and is exactly the unformalized boundary. This note isolates that boundary for expert scrutiny.

1. STATUS AND PURPOSE

This note is a request for a sanity check, not an announcement. It assembles a strengthening of the construction in [FF] from three components:

- (1) a machine-checked analytic theorem: a normal property-(T) subgroup contained in the *compression defect* of a group lies in its MF radical (§2);
- (2) a machine-checked algebraic observation: in the Fournier-Facio configuration the compression defect is exactly the normal closure of the embedded simple factor (§3);
- (3) a small-cancellation step, assembled entirely from published statements of Hull and Osin: the normal closure of the simple factor can be routed *onto* a further quotient while preserving finite presentation, torsion-freeness, two loxodromic generators, and a protected finite set (§4).

Components (1) and (2), the nonsocicity criterion, and every implication downstream of (3) are formalized in Lean 4 in the accompanying repository; §6 states the exact certification boundary. Component (3) is the only part that is not formalized, and it is the part this note asks an expert in acylindrical small cancellation to audit. The main claim is Theorem 5.1.

Throughout, *MF group* has the Carrión–Dadarlat–Eckhardt meaning [CDE]: a countable discrete group G is MF if it embeds in the unitary group of a *norm matrix corona* $\mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$ (bounded sequences modulo sequences vanishing in operator norm). The *MF radical* $\text{Rad}_{\text{MF}}(G)$ is the intersection of the kernels of all homomorphisms $G \rightarrow \mathcal{U}(\mathcal{Q})$ over all coronas; thus G is non-MF as soon as $\text{Rad}_{\text{MF}}(G) \neq 1$, and $\text{Rad}_{\text{MF}}(G) = G$ says that every homomorphism from G to the unitary group of any norm matrix corona is trivial.

2. THE MACHINE-CHECKED OBSTRUCTIONS

Definition 2.1. A *compression core* on a countable group G consists of a group Γ with property (T), a homomorphism $\iota: \Gamma \rightarrow G$, an element $t \in G$ with $t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma)$, and an element $c \in G$ commuting with $\iota(\Gamma)$. Its *defect* is the normal subgroup

$$D = \langle\langle [tct^{-1}, \iota(\gamma)] : \gamma \in \Gamma \rangle\rangle_G.$$

If c still commuted with $\iota(\Gamma)$ *after* transport by t , the defect would vanish; D measures the failure of one-sided Kazhdan transport to be implementable by an honest symmetry.

Theorem 2.2 (machine-checked; `normalKazhdan_le_normMFResidual`). *Let G be a countable group carrying a compression core with defect D , and let $K \trianglelefteq G$ be a normal subgroup with property (T) such that $K \subseteq D$. Then $K \subseteq \text{Rad}_{\text{MF}}(G)$: every homomorphism from G to the unitary group of a norm matrix corona kills K pointwise. No finiteness, centrality, or torsion hypothesis on K is used.*

The second obstruction is the compression nonsoficity criterion, in the form of [FF, Proposition 1.2] (the same mechanism as the criterion driving [OAI, KT]): if $\Gamma \leq G$ are infinite groups with property (T), $G = \langle \Gamma, t_1, \dots, t_m \rangle$ with $t_i \Gamma t_i^{-1} \subseteq \Gamma$, and $J \leq G$ is finitely generated with $[\Gamma, J] = 1$, $\Gamma \cap J = 1$, and $t_1 J t_1^{-1} \subseteq \Gamma$, then soficity of G forces J to be LEF. This criterion is also machine-checked in the repository (`CriterionAssembly.not_isSofic_of_not_isLEF` over the interface `CompressionSetup`). Since a finitely presented LEF group is residually finite (Vershik–Gordon) and an infinite simple group is never residually finite, a finitely presented infinite simple J is never LEF; this is the use made of it below, exactly as in [FF].

3. THE FOURNIER-FACIO CONFIGURATION AND ITS DEFECT

We recall the output of [FF, §2], in the notation of that paper. Fournier-Facio constructs, from a universal finitely presented torsion-free group U , a finitely presented infinite simple torsion-free group S (Burger–Mozes or Hyde–Lodha), and a torsion-free hyperbolic Kazhdan group H :

- (1) a finitely presented torsion-free property-(T) group P containing $P_1 \times P_2 \times S$ with $P_i \cong P$, via small cancellation over the relatively hyperbolic pair $(U * H, U)$ [Os10, Theorem 2.4];
- (2) the double HNN extension $E = \langle P, u_1, u_2 \mid u_i P u_i^{-1} = P_i \rangle$, torsion-free, finitely presented, and acylindrically hyperbolic by [MO];
- (3) a common quotient $\pi: E \twoheadrightarrow G$ of E and H via [Hul, Corollary 7.4], injective on a prescribed finite set containing some $s \in S \setminus \{1\}$; G is finitely presented, torsion-free by [Hul, Theorem 7.1(e)], and has property (T) as a quotient of H .

Write $\Gamma = \pi(P)$, $t_i = \pi(u_i)$, and $J = t_1^{-1} \pi(S) t_1$, so that $[\Gamma, J] = 1$, $\Gamma \cap J = 1$, $t_1 J t_1^{-1} = \pi(S) \subseteq \Gamma$, and $\pi|_S$ is injective by simplicity. (These are the data entering [FF, Proposition 1.2]; the harmless discrepancy between $t_1 J t_1^{-1}$ and $t_1^{-1} J t_1$ conventions is discussed in the companion document cited in §6.)

Remark 3.1 (acylindrical hyperbolicity of G). The paper [FF] does not record whether G is acylindrically hyperbolic, because nothing there needs it. We need it, and it is available at no cost: since E is finitely generated, the proof of [Hul, Corollary 7.4] produces the common quotient in the class $\mathcal{AH}_0$ (two applications of [Hul, Theorem 7.1], the second of which makes the image of the suitable subgroup all of G ; Hull states the $\mathcal{AH}_0$ conclusion explicitly in the

finitely generated case). We therefore fix once and for all a group G produced by running [FF, §2] with this clause of Hull's corollary invoked, so that

G is finitely presented, torsion-free, has property (T), and lies in $\mathcal{AH}_0$.

This is a reading of the cited corollary, not a modification of the construction; it is the first of the two points we would like an expert to confirm.

Now take the compression core on G given by ι the inclusion of Γ , $t = t_1$, and any $c \in J \setminus \{1\}$, so $d := t_1 c t_1^{-1} \in \pi(S) \setminus \{1\}$.

Lemma 3.2 (defect identification). $D = \langle\langle \pi(S) \rangle\rangle_G$. The containment $\supseteq$ is machine-checked, as `simpleSubgroup_le_defectNormal`.

Proof. ($\subseteq$) Each generator $[d, \iota(\gamma)] = d(\gamma d^{-1} \gamma^{-1})$ is a product of two conjugates of $d^{\pm 1} \in \pi(S)$, hence lies in the normal subgroup $\langle\langle \pi(S) \rangle\rangle_G$.

($\supseteq$) $D \cap \pi(S)$ is normal in $\pi(S)$. Since S is infinite simple it is nonabelian with trivial centre, so $d \neq 1$ is not central in $\pi(S)$: there is $\gamma \in \pi(S) \subseteq \Gamma$ with $[d, \gamma] \neq 1$. That commutator lies in $D \cap \pi(S)$, which is therefore nontrivial, hence equals $\pi(S)$ by simplicity; so $\langle\langle \pi(S) \rangle\rangle_G \subseteq D$. $\square$

Thus G itself already has a nonzero defect containing no torsion, and the entire question of applying Theorem 2.2 with K as large as possible reduces to enlarging $\langle\langle \pi(S) \rangle\rangle$. Note that no choice of ingredients makes this automatic: $E/\langle\langle P \rangle\rangle_E$ is free of rank 2, so $\langle\langle S \rangle\rangle_E$ is never all of E , and saturation must be created at a quotient stage.

4. SATURATION VIA HULL'S THEOREM

Everything in this section is assembled from the published statements [Hul, Definition 1.4, Lemma 3.5, Lemma 5.5, Corollary 5.7, Lemma 5.8, Theorem 3.12, Theorem 7.1] and [Os16, Lemma 7.1].

Lemma 4.1. *Let $G \in \mathcal{AH}$ be torsion-free and let $N \neq 1$ be a normal subgroup. Fix a generating set A with $\Gamma(G, A)$ hyperbolic and the G -action acylindrical and non-elementary ([Hul, Theorem 3.12(4)]). Then N is suitable with respect to A in the sense of [Hul, Definition 1.4], and N contains non-commensurable loxodromic elements h_1, h_2 with $E_G(h_i) = \langle h_i \rangle$ such that $S_0 := \langle h_1, h_2 \rangle \leq N$ is again suitable.*

Proof. N is nontrivial and torsion-free, hence infinite; being normal it satisfies $|N^g \cap N| = \infty$ for all g , i.e. it is s -normal, so it acts non-elementarily by [Os16, Lemma 7.1]. Condition (3) of suitability (N normalizes no nontrivial finite subgroup) is vacuous in a torsion-free group, by [Hul, Lemma 5.5]. Now [Hul, Corollary 5.7] applied to the suitable subgroup N yields h_1, h_2 as stated, and [Hul, Lemma 5.8] makes $\langle h_1, h_2 \rangle$ suitable. $\square$

Lemma 4.2 (defect saturation). *Let $G \in \mathcal{AH}$ be finitely presented and torsion-free, let $N \neq 1$ be a normal subgroup, and let $F \subseteq G$ be finite. Then there is a surjection $q: G \twoheadrightarrow Q$ such that*

- (1) $Q \in \mathcal{AH}$;
- (2) Q is finitely presented and torsion-free;
- (3) $q(N) = Q$; indeed $Q = \langle q(h_1), q(h_2) \rangle$ with $h_1, h_2 \in N$ the elements of Lemma 4.1, so Q is 2-generated;
- (4) $q|_F$ is injective.

Proof. Let $S_0 = \langle h_1, h_2 \rangle \leq N$ be as in Lemma 4.1, let $\{t_1, \dots, t_m\}$ be a finite generating set of G , and choose M with $B_A(M) \supseteq F$ (always possible, as Hull notes). Apply [Hul, Theorem 7.1] to (G, S_0) with these t_i and this M : it returns $q: G \twoheadrightarrow Q$ with $Q \in \mathcal{AH}$ (clause (a)), $q|_{B_A(M)}$ injective (clause (b)), and $q(t_i) \in q(S_0)$ for all i (clause (c)). Then

$$Q = \langle q(t_1), \dots, q(t_m) \rangle \subseteq q(S_0) \subseteq q(N) \subseteq Q,$$

so $q(N) = Q = q(S_0) = \langle q(h_1), q(h_2) \rangle$. Torsion-freeness is clause (e). Finite presentation is read off from the proof of Theorem 7.1 exactly as in [FF]: the case $m = 1$ passes to the quotient by the normal closure of a single small-cancellation relator $W = t^{-1}h_1^{m_1}h_2^{l_1} \dots h_1^{m_n}h_2^{l_n}$ (the set of cyclic shifts of $W^{\pm 1}$ has the same normal closure), and the general case is a finite induction; so Q is G with m added relators. $\square$

This is the second point we would like an expert to confirm: that [Hul, Corollary 5.7] may be applied to the suitable subgroup N of Lemma 4.1 (rather than to G itself) and that the resulting $S_0 \leq N$ feeds into [Hul, Theorem 7.1] exactly as Hull's own proof of [Hul, Corollary 7.4] feeds its $S = \langle h_1, h_2 \rangle$. An alternative route with the same output — re-running Hull's proof of Corollary 7.4 with the reservoir of target words taken inside a prescribed infinite normal subgroup of one factor — is written out in the companion document (§6); the route above has the advantage of using Theorem 7.1 as a black box applied to the already-constructed group G .

5. THE MAIN THEOREM

Theorem 5.1. *Let G be the Fournier-Facio group of §3 (with Remark 3.1), and apply Lemma 4.2 with $N = \langle\langle \pi(S) \rangle\rangle_G$ and $F = \{1, \pi(s)\}$, obtaining $q: G \twoheadrightarrow Q$. Then:*

- (1) Q is 2-generated, finitely presented, torsion-free, acylindrically hyperbolic, and has property (T);
- (2) every nontrivial quotient L of Q — including Q itself — satisfies $\text{Rad}_{\text{MF}}(L) = L$; in particular no nontrivial quotient of Q is MF, and every homomorphism from Q to an MF group is trivial;
- (3) every nontrivial quotient of Q is non-sofic.

Proof. (1) All items are Lemma 4.2 except property (T), which passes to the quotient Q of G .

For (2) and (3), let $r: Q \twoheadrightarrow L$ be a nontrivial quotient and write $\rho = r \circ q \circ \pi: E \twoheadrightarrow L$. Since $q(N) = Q$ and images of normal closures are normal closures of images,

$$L = r(Q) = r(\langle\langle q\pi(S) \rangle\rangle_Q) = \langle\langle \rho(S) \rangle\rangle_L.$$

In particular $\rho(S) \neq 1$ (else $L = 1$), so $\rho|_S$ is injective by simplicity and $\rho(S) \cong S$.

The configuration descends. Put $\Gamma_L = \rho(P)$, $t_i^L = \rho(u_i)$, $J_L = (t_1^L)^{-1}\rho(S)t_1^L$. Then $L = \langle \Gamma_L, t_1^L, t_2^L \rangle$; $t_i^L \Gamma_L (t_i^L)^{-1} \subseteq \Gamma_L$; $[\Gamma_L, J_L] = 1$ and $t_1^L J_L (t_1^L)^{-1} = \rho(S) \subseteq \Gamma_L$ (images of the relations in G); Γ_L has property (T) as a quotient of P and is infinite since it contains $\rho(S) \cong S$; and $\Gamma_L \cap J_L = 1$: an element x of the intersection lies in J_L and, by $[\Gamma_L, J_L] = 1$, centralizes J_L , hence lies in the centre of $J_L \cong S$, which is trivial. Finally L has property (T) as a quotient of Q .

(2) Take the compression core on L with ι the inclusion of Γ_L , $t = t_1^L$, $c \in J_L \setminus \{1\}$. By the proof of Lemma 3.2, run verbatim in L (using that $\rho(S) \cong S$ is nonabelian simple with trivial centre), the defect is $\langle\langle \rho(S) \rangle\rangle_L = L$. Apply Theorem 2.2 with $K = L$: $\text{Rad}_{\text{MF}}(L) = L$. Since $L \neq 1$, L is not MF; and a nontrivial homomorphism $Q \rightarrow M$ with M MF would make

its image a nontrivial MF quotient... more precisely, its image is a nontrivial quotient L of Q embedded in M , and subgroups of MF groups are MF, contradicting (2) for L .

(3) The data of the previous paragraph are exactly the hypotheses of [FF, Proposition 1.2] for L , with $J_L \cong S$ finitely generated. If L were sofic, S would be LEF; but S is finitely presented, infinite and simple, hence not LEF (§2). $\square$

Corollary 5.2. *Q has no nontrivial finite, amenable, residually finite, linear, sofic, or MF quotient; Q is perfect; Q has no proper finite-index subgroup; the profinite and Bohr compactifications of Q are trivial; and every finite-dimensional unitary representation of Q is trivial.*

Proof. Finite, amenable, residually finite, and linear groups (Mal'cev) are sofic (or directly MF), so a nontrivial quotient of any of these kinds would contradict Theorem 5.1. An abelian nontrivial quotient would be amenable, so Q is perfect. A proper finite-index subgroup would yield a nontrivial finite quotient of Q (the action on cosets is nontrivial and factors through a finite symmetric group, giving a nontrivial finite image). The compactification and representation statements follow since compact groups are MF-approximable targets in the relevant sense: a nontrivial finite-dimensional unitary image would be a nontrivial linear quotient. $\square$

Corollary 5.3. *Fix a finite presentation $Q = \langle x, y \mid R \rangle$ and a word v with $v \neq 1$ in Q . The set of 2-marked groups satisfying the finitely many relations R and the single inequation $v \neq 1$ is a nonempty clopen subset of the space of 2-marked groups consisting entirely of groups that are simultaneously non-sofic and non-MF, each with full MF radical. Moreover $F_2 \twoheadrightarrow Q$ witnesses that neither soficity nor MF-ness passes to quotients, with a residually finite source; two generators are optimal, since every 1-generated group is abelian, hence MF and sofic.*

Proof. A marked group satisfying R is a marked quotient of Q ; the inequation $v \neq 1$ makes it nontrivial; both conditions are clopen. Apply Theorem 5.1. $\square$

Remark 5.4. The reduced C^* -algebra $C_r^*(Q)$ is stably finite (it has a faithful trace) and not MF, and no unital $*$ -homomorphism $C_{\max}^*(Q) \rightarrow A$ with A an MF algebra can be nontrivial on the canonical unitaries: it would induce a group homomorphism from Q into unitaries liftable to matrix models, factoring through the trivial group by Theorem 5.1. We deliberately do *not* assert simplicity, unique trace, or stable rank one of $C_r^*(Q)$ here: those would follow from the acylindrical-hyperbolicity literature applied to Q (which is torsion-free with trivial finite radical), but the corresponding inputs are outside both the formalized development and the scope of this note.

6. THE CERTIFICATION BOUNDARY

The accompanying Lean 4 development machine-checks the following, with no axioms and no literature-transcription premises, in the modules

`Sofic/NormalKazhdanMFRadical`, `Sofic/TorsionFreeFullMFRadical`,
`Sofic/TorsionFreeFullMFConsequences`, `Criterion/CriterionAssembly`:

- Theorem 2.2 (`normalKazhdan_le_normMFResidual`, also in literal Carrión–Dadarlat–Eckhardt corona form);
- the nonsoficity criterion of §2 (`not_isSofic_of_not_isLEF`);
- the defect identification of Lemma 3.2 from the abstract Fournier-Facio configuration (`FournierFacioDefectData.simpleSubgroup_le_defectNormal`);

- all implications downstream of the saturation step: from proof-carrying input data (`DefectRoutingData`: a surjection with saturated defect, two generators, finite presentation, torsion-freeness in power form, property (T), one protected element) Lean derives the full package of Theorem 5.1(1)–(2) — including `cdeMFRResidual = ⊤`, failure of the CDE embedding property, heredity of the full radical by every quotient, and triviality of every homomorphism to an operator-MF group — and Theorem 5.1(3) for the routed quotient, via `NonsoficCriterionData.not_isSofic` applied to the descended criterion data.

What is *not* formalized, and is asserted here only on the strength of the cited literature, is the existence of the input data: Fournier-Facio’s construction (§3), acylindrical hyperbolicity and the $\mathcal{AH}_0$ clause (Remark 3.1), and Lemmas 4.1–4.2. There is deliberately no Lean theorem of the form “there exists a group Q with these properties”: Mathlib has no theory of acylindrically hyperbolic groups or of Hull’s suitable-subgroup small cancellation, and we have chosen to represent that boundary by structures whose fields are consumed by checked proofs rather than by axioms.

A companion document in the repository,

`docs/TORSION_FREE_NORMAL_GENERATION_HULL_QUOTIENT.md`,

contains the alternative substitution-based route to Lemma 4.2 together with verbatim source quotations of every Hull and Osin statement used above, extracted from the arXiv PDFs on August 14, 2026.

The two questions for the expert reader.

- (1) Remark 3.1: is it correct that running [FF, §2] with the finitely-generated clause of [Hul, Corollary 7.4] yields $G \in \mathcal{AH}_0$ with all other properties of the construction unchanged?
- (2) Lemma 4.2: is the direct application of [Hul, Corollary 5.7 and Theorem 7.1] to the suitable subgroup $N = \langle\langle \pi(S) \rangle\rangle_G$ (suitability via [Os16, Lemma 7.1] and torsion-freeness) correct as stated, including the finite-presentation and torsion-preservation readings of the proof of Theorem 7.1?

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